

Linh Nguyen

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EDUCATION

Florida State University - Tallahassee, FL

May 2027

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 4.0/4.0

Coursework: Data Structure & Algorithms, Parallel & Distributed Programming, Computer Networks, Programming in C++

EXPERIENCE

Google

May 2026 – Aug 2026

Software Engineer Intern

- Automated test data generation for **3,800+ engineers** across **800+ YouTube API endpoints** by building a self-healing agent that dynamically learns implicit entity relationships, breaks down API workflows, and generates **Python** seeding code.
- Mitigated system crash risks by engineering a **secure, stateful REPL sandbox** to verify agent-generated code, utilizing AST-based validation to **block 30+ unsafe operations** with **~2ms runtime overhead**.
- Optimized LLM context-window efficiency by **94%** through a custom I/O interception filter that extracts critical metadata from massive, deeply nested YouTube Protobuf payloads (**up to 1.5MB+**), saving **~100K+ tokens per session**.
- Reduced agent's batch video generation latency by **86%** through engineering a **thread-safe parallel processing pipeline** with a 3-tier cascading fallback mechanism, ensuring **high availability** against AI quota limits.

AIZ (Startup)

Jun 2025 – Aug 2025

Software Engineer Intern

- Engineered an internal assistant chatbot prototype to reduce manual review of **150+ documents** by enabling users to query uploaded PDFs using a **RAG pipeline** with **LangChain** and **OpenAI LLMs**, achieving **91% answer accuracy**.
- Reduced multi-document query latency by **55%** through designing an **in-memory caching layer** for vector indexes and retrievers, eliminating redundant index reloading and rebuilding across repeated queries within a session.
- Built an end-to-end document processing pipeline to accelerate document understanding by extracting and summarizing PDF content using **PyMuPDF** and **Tesseract OCR**, containerized with **Docker** and deployed on **AWS EC2**.

Florida State University

Sept 2024 – May 2026

Artificial Intelligence Research Assistant | [Research paper](#)

- Productionized a co-authored research paper by engineering a cross-platform, **multi-LLM** behavioral health app that enhances conversation continuity and proactive engagement using **React Native**, **Flask API**, and **SQLite**.
- Built an end-to-end machine learning pipeline using **Python**, **Pandas**, **NumPy**, and **Scikit-learn** to predict GPU combined inference cost, achieving **97%+ accuracy** through feature engineering, dataset curation, and model tuning.
- Reviewed **100+ research papers** on Explainable AI for cybersecurity, analyzing techniques, limitations, and tradeoffs in adversarial robustness and threat interpretability to produce a **survey paper** identifying research gaps and future directions.

FELLOWSHIP/PROJECT

Viettech Mentorship - FairShare | [Github](#)

Mar 2025 – July 2025

- Collaborated with a cross-functional team of 4 software engineers and 1 product designer to develop an **iOS app** for sharing and managing group subscriptions with recurring payments using **React Native (Expo)**, **Stripe API**, **Clerk**, and **MongoDB**.
- Led the technical design and implementation of multi-user, multi-group expense management logic, architecting backend APIs and data models for group creation, invitations, and membership management using **Node.js**, **MongoDB**, and **Prisma**.
- Ensured data consistency across multi-user expenses by architecting idempotent recalculation logic for **8+ complex edge cases** (e.g., mid-cycle membership changes), eliminating concurrency risks during multi-step updates.

Cornell Breakthrough Tech AI | [Github](#)

May 2025 – Apr 2026

- Led a team of 3 students to build an end-to-end **Machine Learning ensembling pipeline** to predict wildfire time-to-threat for evacuation zones, with **95% accuracy** in the WiDS Global Datathon using feature engineering and cross-validation.
- Developed a machine learning pipeline to predict Life Ladder scores using the World Happiness dataset, achieving **93% accuracy** by applying data cleaning, feature engineering, and Random Forest optimization.
- Completed a 3-month hands-on machine learning training, implementing end-to-end ML workflows (data preprocessing, model training, evaluation, and cross-validation) on real-world datasets using **Python**, **NumPy**, **Pandas**, and **Scikit-learn**.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, HTML/CSS, TypeScript

Frameworks: gRPC, LangChain, Expo, React.js, Angular.js, Spring Boot, Django, Node.js, Next.js, React Native, Flask API

Libraries & Tools: AWS, NumPy, Matplotlib, Seaborn, Git, Docker, MySQL, MongoDB, PostgreSQL, SQLite, Figma, UNIX, MATLAB, Clerk, OpenAI, PyMuPDF, Tesseract OCR, GoogleAuth, Protobuf